ChatGPT said:

1,000 Hard Skills for Human Jobs on Planet Earth

A **hard skill** is a specific, teachable, and measurable ability that a person can learn through education, training, practice, or work experience. Hard skills help people perform professional tasks, operate tools, solve technical problems, produce goods, and deliver services.

For your Human Skills learning model, we can organize 1,000 hard skills into **50 professional categories, with 20 skills in each category**. This structure covers traditional jobs, modern technology, business, science, nature, and emerging industries.

The 50 categories of hard skills

1,000 hard skills

50 categories × 20 skills per category

Human job learning model

From basic work skills to advanced professional and technical skills

Part 1 — Business, management, and commerce

1. Accounting and bookkeeping

1. Recording financial transactions
2. Double-entry bookkeeping
3. General ledger management
4. Accounts payable processing
5. Accounts receivable processing
6. Bank reconciliation
7. Cash reconciliation
8. Invoice preparation
9. Receipt management
10. Expense classification
11. Payroll accounting
12. Fixed asset accounting
13. Depreciation calculation
14. Inventory accounting
15. Cost accounting
16. Financial statement preparation
17. Trial balance preparation

18. Tax accounting
19. Accounting software operation
20. Month-end closing

2. Finance and financial analysis

21. Financial planning
22. Budget preparation
23. Cash flow forecasting
24. Financial modeling
25. Financial ratio analysis
26. Profitability analysis
27. Break-even analysis
28. Capital budgeting
29. Investment appraisal
30. Working capital management
31. Debt analysis
32. Equity valuation
33. Business valuation
34. Credit risk analysis
35. Financial risk modeling
36. Treasury management
37. Cost of capital calculation
38. Discounted cash flow valuation
39. Financial scenario analysis
40. Financial performance reporting

3. Business administration and operations

41. Business process mapping
42. Standard operating procedure writing
43. Document control
44. Records management
45. Office administration
46. Business correspondence
47. Meeting documentation
48. Workflow design
49. Resource allocation
50. Operational reporting
51. Business policy documentation
52. Procurement administration

53. Vendor database management
54. Contract administration
55. Business continuity planning
56. Quality procedure implementation
57. Service-level agreement monitoring
58. Administrative database management
59. Business process improvement
60. Operations dashboard preparation

4. Human resources and recruitment

61. Job description writing
62. Job specification development
63. Resume screening
64. Applicant tracking system operation
65. Interview scorecard design
66. Competency assessment
67. Employee onboarding administration
68. Employee record management
69. Payroll data administration
70. Attendance system management
71. Leave administration
72. Employee performance appraisal design
73. Training needs analysis
74. Training program development
75. Learning management system administration
76. Compensation benchmarking
77. Benefits administration
78. Workforce planning
79. HR analytics
80. HR information system operation

5. Project management

81. Project charter development
82. Work breakdown structure creation
83. Project scheduling
84. Gantt chart development
85. Critical path analysis
86. Project budgeting
87. Project cost tracking

88. Resource scheduling
89. Risk register development
90. Project scope management
91. Change request management
92. Milestone tracking
93. Project status reporting
94. Earned value analysis
95. Stakeholder register maintenance
96. Project documentation
97. Agile sprint planning
98. Scrum board management
99. Project portfolio reporting
100. Project closure documentation

6. Strategic management and business development

101. Market sizing
102. Industry analysis
103. Competitor benchmarking
104. SWOT analysis
105. PESTLE analysis
106. Business model design
107. Business plan development
108. Strategic planning
109. Business case preparation
110. Feasibility study preparation
111. Revenue model design
112. Partnership agreement development
113. Franchise model analysis
114. Licensing model analysis
115. Business expansion planning
116. Market entry analysis
117. Scenario planning
118. Corporate performance measurement
119. Balanced scorecard development
120. Strategic KPI design

7. Sales and business development

121. Lead database management
122. Lead qualification

123. Sales pipeline management
124. CRM software operation
125. Sales forecasting
126. Sales target allocation
127. Sales territory mapping
128. Sales quotation preparation
129. Proposal writing
130. Product demonstration
131. Sales order processing
132. Sales contract administration
133. Key account management
134. Dealer network management
135. Distributor relationship administration
136. Sales commission calculation
137. Sales performance analysis
138. Customer segmentation
139. Upselling and cross-selling techniques
140. Sales funnel analysis

8. Marketing and advertising

141. Marketing plan development
142. Market research
143. Customer persona development
144. Brand positioning
145. Marketing campaign planning
146. Advertising budget management
147. Search engine optimization
148. Search engine marketing
149. Email campaign automation
150. Social media campaign management
151. Content calendar development
152. Marketing copywriting
153. Marketing analytics
154. Conversion rate optimization
155. A/B testing
156. Customer acquisition cost calculation
157. Return on advertising spend calculation
158. Marketing attribution analysis
159. Affiliate marketing management
160. Marketing automation platform operation

9. Retail, wholesale, and distribution

161. Retail merchandising
162. Wholesale order management
163. Product assortment planning
164. Shelf space allocation
165. Point-of-sale system operation
166. Retail pricing
167. Promotional pricing
168. Stock replenishment planning
169. Dealer onboarding
170. Distributor territory allocation
171. Trade promotion administration
172. Consignment inventory management
173. Retail sales reporting
174. Wholesale margin calculation
175. Distribution route planning
176. Channel performance analysis
177. Store layout planning
178. Retail shrinkage analysis
179. Returns processing
180. Omnichannel order management

10. Procurement and supply chain

181. Purchase requisition processing
182. Request for quotation preparation
183. Supplier sourcing
184. Supplier quotation comparison
185. Supplier evaluation
186. Purchase order administration
187. Contract purchasing
188. Strategic sourcing
189. Procurement cost analysis
190. Supplier performance measurement
191. Supply chain mapping
192. Demand forecasting
193. Supply planning
194. Material requirements planning
195. Inventory optimization

196. Safety stock calculation
197. Reorder point calculation
198. Logistics cost analysis
199. Procurement compliance auditing
200. Supply chain risk assessment

Part 2 — Customer service, information, and technology

11. Customer service and customer relationship management

201. Customer database administration
202. Customer ticket management
203. Contact center software operation
204. Help desk system operation
205. Customer complaint classification
206. Service request processing
207. Customer feedback analysis
208. Customer satisfaction survey design
209. Net Promoter Score calculation
210. Customer retention analysis
211. Customer journey mapping
212. Customer service quality auditing
213. Warranty claim administration
214. Product return authorization
215. Service level monitoring
216. Customer knowledge base development
217. Chat support platform operation
218. Customer escalation tracking
219. Customer lifetime value calculation
220. Customer relationship reporting

12. Data analysis and statistics

221. Data collection
222. Data cleaning
223. Data validation
224. Spreadsheet data analysis
225. Descriptive statistics
226. Probability calculation
227. Statistical hypothesis testing
228. Correlation analysis
229. Regression analysis

230. Sampling design
231. Survey data analysis
232. Time series analysis
233. Statistical process control
234. Experimental design
235. Data normalization
236. Data visualization
237. Dashboard development
238. Predictive analytics
239. Data quality assessment
240. Statistical software operation

13. Information technology support

241. Computer hardware installation
242. Operating system installation
243. Operating system configuration
244. Software installation
245. Software troubleshooting
246. Computer maintenance
247. Printer configuration
248. Peripheral device configuration
249. User account administration
250. Password policy configuration
251. Desktop support
252. Remote technical support
253. IT asset inventory
254. IT service ticket processing
255. Software patch management
256. Device driver installation
257. File permission configuration
258. Endpoint device configuration
259. Technical documentation
260. IT incident classification

14. Software development

261. Programming logic implementation
262. Algorithm design
263. Data structure implementation
264. Python programming

265. Java programming
266. C++ programming
267. C# programming
268. JavaScript programming
269. TypeScript programming
270. PHP programming
271. Go programming
272. Rust programming
273. Object-oriented programming
274. Functional programming
275. API development
276. Unit testing
277. Integration testing
278. Code debugging
279. Version control with Git
280. Software documentation

15. Web development and digital products

281. HTML authoring
282. CSS styling
283. Responsive web design
284. Front-end development
285. Back-end development
286. Full-stack development
287. Web server configuration
288. Domain and DNS configuration
289. Website deployment
290. Content management system administration
291. E-commerce website configuration
292. Payment gateway integration
293. Web accessibility testing
294. Website performance optimization
295. Web analytics configuration
296. Progressive web app development
297. Mobile web optimization
298. Web application security testing
299. Search engine indexing configuration
300. Website backup and restoration

16. Databases and data engineering

301. Relational database design
302. SQL querying
303. Database normalization
304. Database indexing
305. Database administration
306. Database backup management
307. Database recovery
308. Database performance tuning
309. Stored procedure development
310. Data warehouse design
311. Data lake architecture
312. ETL pipeline development
313. ELT pipeline development
314. Data integration
315. Data migration
316. Master data management
317. Metadata management
318. Data catalog administration
319. Stream processing
320. Data pipeline monitoring

17. Artificial intelligence and machine learning

321. Machine learning data preparation
322. Feature engineering
323. Supervised learning model development
324. Unsupervised learning model development
325. Classification model training
326. Regression model training
327. Clustering analysis
328. Neural network development
329. Deep learning model training
330. Natural language processing
331. Computer vision model development
332. Model evaluation
333. Model validation
334. Hyperparameter tuning
335. Model deployment
336. MLOps pipeline development
337. Retrieval-augmented generation implementation

338. Large language model API integration
339. AI output evaluation
340. AI model monitoring

18. Cybersecurity and information security

341. Security risk assessment
342. Network security configuration
343. Firewall administration
344. Identity and access management
345. Multi-factor authentication deployment
346. Security log analysis
347. Vulnerability scanning
348. Penetration testing
349. Security patch management
350. Endpoint detection system administration
351. Intrusion detection configuration
352. Incident response execution
353. Digital forensics
354. Malware analysis
355. Encryption implementation
356. Public key infrastructure administration
357. Security audit execution
358. Security awareness training development
359. Data loss prevention configuration
360. Security compliance assessment

19. Cloud computing and DevOps

361. Cloud infrastructure provisioning
362. Virtual machine administration
363. Container image development
364. Docker configuration
365. Kubernetes administration
366. Infrastructure as code
367. Continuous integration pipeline development
368. Continuous delivery pipeline development
369. Automated deployment
370. Cloud storage configuration
371. Cloud network configuration
372. Cloud identity management

373. Cloud cost optimization
374. Serverless application development
375. System observability configuration
376. Log aggregation
377. Performance monitoring
378. Disaster recovery configuration
379. Configuration management
380. Site reliability engineering

20. Enterprise software and business systems

381. ERP system configuration
382. ERPNext administration
383. SAP system operation
384. Oracle enterprise application operation
385. Microsoft Dynamics configuration
386. CRM system implementation
387. HRIS configuration
388. Inventory management system configuration
389. Warehouse management system configuration
390. Point-of-sale system implementation
391. Accounting system configuration
392. Workflow automation
393. Business rules configuration
394. Enterprise application integration
395. User acceptance testing
396. Business requirements documentation
397. System implementation planning
398. Data import and export
399. Enterprise reporting configuration
400. Software license management

Part 3 — Design, media, and communication production

21. Graphic design and visual communication

401. Typography selection
402. Color theory application
403. Logo design
404. Brand identity system design
405. Layout composition
406. Vector illustration

407. Raster image editing
408. Image retouching
409. Icon design
410. Infographic design
411. Poster design
412. Banner design
413. Brochure design
414. Packaging graphics design
415. Print-ready artwork preparation
416. Prepress file preparation
417. Digital advertisement design
418. Design system development
419. Visual hierarchy construction
420. Brand guideline documentation

22. UI/UX and product design

421. User research interview design
422. User survey design
423. User persona development
424. User journey mapping
425. Information architecture
426. User flow diagramming
427. Wireframing
428. Interactive prototyping
429. Interface layout design
430. Usability testing
431. Accessibility evaluation
432. Interaction design
433. Design system component development
434. Design token management
435. Mobile app interface design
436. Responsive interface design
437. User behavior analytics
438. Conversion-oriented interface testing
439. Design handoff documentation
440. Product requirements specification

23. Photography and image production

441. Camera operation

142. Exposure control
143. Aperture selection
144. Shutter speed control
145. ISO setting
146. Lens selection
147. Studio lighting setup
148. Portrait photography
149. Product photography
150. Architectural photography
151. Food photography
152. Landscape photography
153. Macro photography
154. Event photography
155. Image composition
156. White balance correction
157. RAW image processing
158. Photo color grading
159. Image cataloging
160. Photographic print preparation

24. Video production and animation

161. Video camera operation
162. Shot planning
163. Storyboard development
164. Video lighting setup
165. Audio recording
166. Video editing
167. Timeline editing
168. Color correction
169. Color grading
170. Motion graphics creation
171. 2D animation
172. 3D animation
173. Character rigging
174. Visual effects compositing
175. Green screen keying
176. Audio synchronization
177. Subtitle production
178. Video compression
179. Live streaming setup

480. Video delivery specification management

25. Writing, publishing, and translation

481. Technical writing
482. Instruction manual writing
483. Standard operating procedure writing
484. Business report writing
485. Proposal writing
486. Grant application writing
487. Copyediting
488. Proofreading
489. Copywriting
490. Search-optimized content writing
491. Scriptwriting
492. Editorial planning
493. Fact-checking
494. Citation management
495. Translation
496. Localization
497. Subtitling
498. Transcription
499. Terminology management
500. Document formatting

Part 4 — Engineering, construction, and manufacturing

26. Architecture and building design

501. Architectural drafting
502. Building code interpretation
503. Site analysis
504. Space planning
505. Floor plan development
506. Elevation drawing
507. Building section drawing
508. Building information modeling
509. Revit modeling
510. AutoCAD drafting
511. SketchUp modeling
512. Architectural visualization
513. Building material specification

514. Building envelope design
515. Accessibility design
516. Sustainable building design
517. Construction drawing coordination
518. Area and quantity calculation
519. Design specification writing
520. Building permit documentation

27. Civil and structural engineering

521. Land surveying
522. Topographic mapping
523. Site grading design
524. Road alignment design
525. Drainage system design
526. Stormwater management
527. Water supply network design
528. Sewer network design
529. Foundation design
530. Structural load calculation
531. Reinforced concrete design
532. Steel structure design
533. Geotechnical investigation
534. Soil bearing capacity analysis
535. Retaining wall design
536. Bridge engineering
537. Pavement design
538. Construction material testing
539. Structural inspection
540. Infrastructure cost estimation

28. Mechanical engineering

541. Mechanical component design
542. Engineering drawing interpretation
543. Tolerance specification
544. Geometric dimensioning and tolerancing
545. Machine element calculation
546. Gear design
547. Bearing selection
548. Shaft design

549. Fluid mechanics calculation
550. Thermodynamics analysis
551. Heat transfer calculation
552. HVAC system design
553. Pump selection
554. Hydraulic system design
555. Pneumatic system design
556. Mechanical stress analysis
557. Finite element analysis
558. Computer-aided engineering
559. Mechanical prototype testing
560. Equipment performance evaluation

29. Electrical and electronic engineering

561. Electrical circuit analysis
562. Electrical schematic drafting
563. Electrical wiring installation
564. Electrical load calculation
565. Power distribution design
566. Transformer selection
567. Motor control configuration
568. Programmable logic controller programming
569. Industrial sensor integration
570. Printed circuit board design
571. Electronic circuit simulation
572. Microcontroller programming
573. Embedded system development
574. Signal processing
575. Power electronics design
576. Electrical grounding design
577. Electrical protection coordination
578. Electrical safety testing
579. Instrument calibration
580. Electrical fault diagnosis

30. Construction management and quantity surveying

581. Construction work scheduling
582. Construction cost estimation
583. Bill of quantities preparation

584. Material quantity takeoff
585. Construction tender preparation
586. Contractor evaluation
587. Construction procurement planning
588. Site progress measurement
589. Variation order assessment
590. Interim payment certification
591. Construction contract administration
592. Subcontractor coordination documentation
593. Construction quality inspection
594. Site safety inspection
595. As-built drawing preparation
596. Construction claims documentation
597. Project cash flow forecasting
598. Construction resource planning
599. Defect inspection reporting
600. Project handover documentation

31. Skilled trades and building services

601. Bricklaying
602. Concrete mixing and placement
603. Concrete formwork installation
604. Rebar cutting and fixing
605. Tile installation
606. Stone installation
607. Plastering
608. Painting and surface finishing
609. Carpentry
610. Cabinet installation
611. Roof installation
612. Waterproofing application
613. Plumbing pipe installation
614. Sanitary fixture installation
615. Drainage pipe installation
616. Welding
617. Metal fabrication
618. Electrical fixture installation
619. Air conditioning installation
620. Building maintenance repair

32. Manufacturing and industrial production

521. Production process planning
522. Production scheduling
523. Machine setup
524. Machine operation
525. Assembly line operation
526. Production capacity calculation
527. Manufacturing cost calculation
528. Work instruction preparation
529. First article inspection
530. Statistical process control
531. Lean manufacturing implementation
532. Six Sigma methodology
533. Root cause analysis
534. Corrective action implementation
535. Preventive maintenance planning
536. Overall equipment effectiveness calculation
537. Production yield analysis
538. Scrap rate analysis
539. Factory layout planning
540. Production traceability management

33. Industrial automation and robotics

541. Industrial robot programming
542. Robot path planning
543. Robotic arm calibration
544. PLC ladder logic programming
545. Human-machine interface configuration
546. Supervisory control system configuration
547. Industrial network configuration
548. Machine vision integration
549. Automated inspection setup
550. Conveyor automation design
551. Automated guided vehicle configuration
552. Autonomous mobile robot deployment
553. Servo motor tuning
554. Motion control programming
555. Safety interlock configuration
556. Digital twin development

657. Industrial sensor calibration
658. Robotics simulation
659. Automated material handling design
660. Automation system troubleshooting

34. Quality management and compliance

661. Quality management system implementation
662. ISO 9001 implementation
663. Quality audit execution
664. Inspection plan development
665. Sampling inspection
666. Measurement system analysis
667. Gauge repeatability and reproducibility analysis
668. Nonconformance reporting
669. Corrective and preventive action management
670. Failure mode and effects analysis
671. Control plan development
672. Supplier quality auditing
673. Product specification verification
674. Calibration program management
675. Documented procedure control
676. Regulatory compliance mapping
677. Product certification administration
678. Traceability system implementation
679. Customer quality complaint investigation
680. Quality performance dashboard development

35. Automotive, machinery, and transportation equipment

681. Vehicle engine diagnostics
682. Vehicle electrical diagnostics
683. Vehicle brake servicing
684. Vehicle suspension servicing
685. Transmission maintenance
686. Tire inspection and replacement
687. Wheel alignment
688. Diesel engine maintenance
689. Heavy equipment operation
690. Excavator operation
691. Crane operation

692. Forklift operation
693. Truck maintenance
694. Fleet maintenance planning
695. Vehicle inspection
696. Hydraulic equipment repair
697. Automotive air conditioning servicing
698. Electric vehicle battery diagnostics
699. Electric motorbike maintenance
700. Vehicle telematics system configuration

Part 5 — Science, nature, agriculture, and the environment

36. Biology and life sciences

701. Biological specimen preparation
702. Microscopy
703. Cell culture techniques
704. DNA extraction
705. Polymerase chain reaction testing
706. DNA sequencing analysis
707. Protein analysis
708. Enzyme activity measurement
709. Microbial culture techniques
710. Microbial identification
711. Histological sample preparation
712. Flow cytometry
713. Genetic data analysis
714. Bioinformatics pipeline development
715. Ecological field sampling
716. Species identification
717. Taxonomic classification
718. Population sampling
719. Laboratory experiment design
720. Biological data interpretation

37. Chemistry and materials science

721. Chemical solution preparation
722. Analytical balance operation
723. Titration
724. pH measurement
725. Spectrophotometry

726. Chromatography
727. Mass spectrometry
728. Chemical reaction analysis
729. Chemical synthesis
730. Chemical purity testing
731. Polymer characterization
732. Metal composition analysis
733. Material hardness testing
734. Tensile strength testing
735. Thermal analysis
736. Corrosion testing
737. Cement composition analysis
738. Ceramic material testing
739. Paint formulation testing
740. Laboratory chemical waste handling

38. Physics and applied mathematics

741. Classical mechanics calculation
742. Electromagnetic field analysis
743. Optics measurement
744. Thermodynamic modeling
745. Quantum mechanics calculation
746. Numerical methods
747. Differential equation solving
748. Linear algebra computation
749. Mathematical optimization
750. Operations research modeling
751. Monte Carlo simulation
752. Computational fluid dynamics
753. Finite element simulation
754. Experimental physics measurement
755. Uncertainty analysis
756. Dimensional analysis
757. Mathematical modeling
758. Signal analysis
759. Scientific computing
760. Physics laboratory instrumentation

39. Agriculture and crop production

761. Soil sampling
762. Soil fertility assessment
763. Seed selection
764. Seed germination testing
765. Land preparation
766. Crop planting
767. Irrigation scheduling
768. Fertilizer application
769. Pest identification
770. Crop disease diagnosis
771. Weed management
772. Integrated pest management
773. Crop rotation planning
774. Greenhouse management
775. Hydroponic system operation
776. Crop yield estimation
777. Harvest scheduling
778. Postharvest handling
779. Agricultural machinery operation
780. Precision agriculture system operation

40. Forestry, wildlife, and biodiversity

781. Tree species identification
782. Forest inventory measurement
783. Tree diameter measurement
784. Forest mapping
785. Timber volume estimation
786. Forest regeneration planning
787. Nursery seedling production
788. Reforestation planning
789. Forest fire risk mapping
790. Wildlife population surveying
791. Wildlife camera trap monitoring
792. Biodiversity survey design
793. Habitat mapping
794. Ecosystem restoration planning
795. Invasive species monitoring
796. Fisheries population assessment
797. Aquatic habitat assessment
798. Protected area monitoring

799. Conservation data management
800. Environmental DNA sampling

41. Environmental science and sustainability

801. Environmental impact assessment
802. Environmental baseline surveying
803. Air quality monitoring
804. Water quality testing
805. Soil contamination testing
806. Noise pollution measurement
807. Waste characterization
808. Wastewater sampling
809. Greenhouse gas inventory preparation
810. Carbon footprint calculation
811. Life cycle assessment
812. Environmental management system implementation
813. Environmental compliance auditing
814. Renewable energy resource assessment
815. Energy efficiency auditing
816. Water resource management
817. Circular economy modeling
818. Sustainability indicator reporting
819. Climate risk assessment
820. Environmental remediation planning

Part 6 — Health, food, and human well-being

42. Medicine and clinical laboratory science

821. Clinical specimen collection
822. Blood sample processing
823. Hematology testing
824. Clinical chemistry testing
825. Medical microbiology testing
826. Urinalysis
827. Blood typing
828. Immunoassay testing
829. Clinical laboratory quality control
830. Medical equipment calibration
831. Electrocardiogram recording
832. Medical imaging equipment operation

333. Radiographic image acquisition
334. Ultrasound image acquisition
335. Clinical data entry
336. Medical coding
337. Medical record system operation
338. Laboratory result verification
339. Infection control procedure implementation
340. Clinical research data management

43. Nursing, rehabilitation, and allied health

341. Vital signs measurement
342. Patient admission documentation
343. Medication administration documentation
344. Wound dressing
345. Sterile instrument preparation
346. Specimen collection for clinical testing
347. Patient mobility assistance techniques
348. Basic life support certification skills
349. Cardiopulmonary resuscitation
350. Oxygen equipment operation
351. Patient monitoring equipment operation
352. Physical therapy exercise implementation
353. Range-of-motion assessment
354. Gait assessment
355. Assistive device fitting
356. Occupational therapy activity assessment
357. Speech-language assessment techniques
358. Audiometry testing
359. Vision screening
360. Rehabilitation progress documentation

44. Pharmacy and pharmaceutical science

361. Prescription data processing
362. Medication inventory management
363. Pharmaceutical compounding
364. Dosage calculation
365. Medication labeling
366. Medication storage monitoring
367. Cold-chain medication handling

368. Pharmaceutical stability testing
369. Drug dissolution testing
370. Pharmaceutical quality control
371. Drug interaction database use
372. Pharmacovigilance data processing
373. Clinical trial medication accountability
374. Pharmaceutical manufacturing documentation
375. Good manufacturing practice documentation
376. Pharmaceutical packaging inspection
377. Medicine batch traceability
378. Controlled temperature monitoring
379. Pharmaceutical regulatory documentation
380. Medication dispensing system operation

45. Food science, nutrition, and hospitality

381. Food safety inspection
382. Hazard analysis and critical control point implementation
383. Food microbiological testing
384. Food composition analysis
385. Nutrition calculation
386. Menu nutritional analysis
387. Recipe standardization
388. Food portion control
389. Commercial kitchen equipment operation
390. Food preparation techniques
391. Baking and pastry production
392. Beverage preparation
393. Food preservation
394. Food packaging testing
395. Shelf-life testing
396. Food product development
397. Restaurant inventory control
398. Hotel property management system operation
399. Room reservation system administration
400. Food service cost calculation

Part 7 — Logistics, public infrastructure, and specialist services

46. Logistics, warehousing, and international trade

401. Warehouse layout planning

902. Goods receiving inspection
903. Put-away location assignment
904. Bin location management
905. Picking list processing
906. Packing specification implementation
907. Barcode system operation
908. RFID inventory tracking
909. Cycle counting
910. Warehouse stock reconciliation
911. Freight booking
912. Shipment tracking
913. Cargo load planning
914. Container utilization calculation
915. Customs declaration preparation
916. Import documentation preparation
917. Export documentation preparation
918. Incoterms application
919. Freight cost calculation
920. Delivery route optimization

47. Education, training, and research

921. Curriculum mapping
922. Lesson plan development
923. Learning objective formulation
924. Instructional material development
925. Assessment rubric design
926. Examination question development
927. Student information system operation
928. Learning management system configuration
929. Educational data analysis
930. Competency-based assessment design
931. Training needs assessment
932. Training evaluation design
933. E-learning course development
934. Educational video production
935. Laboratory teaching preparation
936. Research question formulation
937. Literature review methodology
938. Research data collection
939. Research ethics documentation

940. Academic manuscript formatting

48. Law, governance, and public administration

941. Legal document drafting
942. Contract review
943. Legal research
944. Statutory interpretation
945. Regulatory compliance monitoring
946. Corporate governance documentation
947. Company registration processing
948. Business license application processing
949. Tax compliance documentation
950. Intellectual property registration
951. Trademark search
952. Patent database research
953. Procurement tender evaluation
954. Public budget analysis
955. Policy impact assessment
956. Government records management
957. Land title document review
958. Immigration document processing
959. Administrative case management
960. Public service performance reporting

49. Energy, utilities, and telecommunications

961. Solar photovoltaic system design
962. Solar panel installation
963. Solar inverter configuration
964. Battery energy storage system monitoring
965. Wind turbine maintenance
966. Hydropower equipment inspection
967. Electrical grid load forecasting
968. Power generation performance monitoring
969. Energy meter installation
970. Energy consumption auditing
971. Water treatment plant operation
972. Wastewater treatment plant operation
973. Water distribution network maintenance
974. Gas pipeline inspection

975. Fiber optic cable splicing
976. Telecommunications network configuration
977. Radio frequency measurement
978. Mobile network performance testing
979. Satellite communication equipment operation
980. Internet network troubleshooting

50. Emerging technology and specialized professional skills

981. Geographic information system mapping
982. Remote sensing image analysis
983. Drone flight planning
984. Drone mapping and surveying
985. 3D scanning
986. 3D computer-aided design modeling
987. Additive manufacturing setup
988. 3D printing process optimization
989. Virtual reality application development
990. Augmented reality application development
991. Blockchain application development
992. Smart contract programming
993. Internet of Things device integration
994. Digital twin system implementation
995. Robotic process automation development
996. Low-code application development
997. No-code workflow configuration
998. Digital identity system administration
999. Privacy impact assessment
000. Technology implementation evaluation

How to use these 1,000 hard skills

For your Human Job learning model, each skill can be developed using a consistent formula:

Hard Skill Learning Formula

1. Knowledge
Understand the concepts, rules, and principles.

↓

2. Tools and Methods

Learn the software, equipment, techniques, and procedures.

↓

3. Practice

Perform real tasks and solve practical problems.

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4. Measurable Results

Demonstrate accuracy, quality, speed, safety, and productivity.

Recommended professional skill record

You can turn every skill into a record in your Human Job database or learning system.

Field	Example
Skill ID	HS-001
Skill category	Accounting
Skill name	Bank Reconciliation
Skill definition	Matching bank transactions against accounting records
Required knowledge	Bank statements, ledger entries, reconciliation rules
Tools	Spreadsheet or accounting software
Practical task	Reconcile one month's bank transactions
Assessment method	Accuracy of the reconciliation
Skill level	Beginner, Intermediate, Advanced, Expert
Job applications	Accountant, Bookkeeper, Finance Officer

Important distinction: these 1,000 items are a structured starting taxonomy, not a universal official list of every hard skill. Some skills overlap, and certain occupations require licenses, certifications, supervised training, or additional safety qualifications.

The next step for a complete Human Job model is to connect these hard skills to **job titles, soft skills, required qualifications, practical assessments, and career development paths**. This would transform the list into a usable professional skills framework.